# Metatime Cosmology: Lambda–Plasma Emergence, Topological Time-Loops and Falsifiable Predictions

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**Abstract**

We present a formal cosmological model built from the Metatime framework: a tensor–scalar metatime field, closed Collatz-like time-loops (topological cycles), an Euler–Berry global constraint, and nonlocal couplings ("white threads"). The paper states the model assumptions, derives a family-wise mass mapping, solves the fitting problem explicitly, performs the neutrino splitting test, and presents cosmological consequences in a falsifiable form. All numeric solves use the eigenvalue assignments and representative particle masses defined in the project files; outputs and residuals are given explicitly and collected in landscape tables. This manuscript is a specification and calculation package, not an empirical claim that the model is physically realized.

## Contents

**1 Introduction and scope** **2**

**2 Formal model — core definitions** **2**
2.1 Metatime field and manifold . . . 2
2.2 Time operator and eigenvalues . . . 2
2.3 Topological constraints: Berry phase and Euler-Berry condition . . . 3
2.4 Mass operator (ansatz) . . . 3
2.5 White threads and network states . . . 3
2.6 Effective cosmological term . . . 3

**3 Family-wise fitting methodology** **4**
3.1 Power-law fit via log transform (ordinary least squares) . . . 4
3.2 Polynomial fit for mass-squared (exact solve) . . . 4

**4 Data used (project assignments)** **4**
4.1 Eigenvalues $\tau_i$ (project assignment) . . . 4
4.2 Representative observed masses (used for fitting) . . . 5

**5 Solving the fits — explicit algebra and numeric results** **5**
5.1 Power-law fits (log-OLS) per family . . . 5
5.2 Polynomial fit for charged leptons (exact solve) . . . 5
5.3 Neutrino splitting test (pairwise scaling) . . . 6

**6 Emergence of Discrete Mass Spectra via Topological Time-Loops** **6**

**7 Geometric Quantization of Shells via the Metatime Lagrangian** **7**

**8 Cosmological consequences and falsifiable targets** **7**

**9 Discussion and model limitations** **8**

**10 Conclusion** **8**

**A Reproducibility notes** **8**

**B Numeric solutions and full tables** **8**

# 1 Introduction and scope

This document formalizes a cosmological model derived from the Metatime project files. The goal is to produce a mathematically explicit, numerically reproducible, and falsifiable model connecting:

- a metatime tensor–scalar field $\mathcal{T} = (\phi, T_{\mu\nu})$ on a Kähler manifold,
- discrete topological cycles (Collatz-like loops) with integer lengths $\tau_i$,
- a mass operator $\hat{\mathcal{M}}(\tau)$ whose eigenvalues map to particle masses,
- a global Euler–Berry constraint that discretizes admissible states,
- white-thread couplings $\mathcal{W}_{ij}$ mediating nonlocal correlations,
- a dynamic, topology-modulated cosmological term $\Lambda_{\text{eff}}(x)$.

We give formal definitions (Section 2), derive the fitting methodology (Section 3), solve the numerical fits and the neutrino test (Section 5), and summarize falsifiable cosmological predictions (Section 8). The Appendix contains the full numeric tables set in landscape orientation.

# 2 Formal model — core definitions

## 2.1 Metatime field and manifold

We work on a metatime configuration manifold $M_{\text{time}}$ (Kähler-type in the project files). The metatime field is a paired object

$$\mathcal{T}(x) = \big(\phi(x),\ T_{\mu\nu}(x)\big),$$

where $\phi$ is a scalar degree of freedom and $T_{\mu\nu}$ a symmetric tensor field encoding local metatime energy/flux.

A path (cycle) in $M_{\text{time}}$ is parametrized by an internal variable $\tau$; closed cycles are labeled by an integer cycle length $\tau_{\text{cycle}} \in \mathbb{Z}^+$ (the "Collatz length" assignment in the project).

## 2.2 Time operator and eigenvalues

Define a metatime time-operator $\hat{T}(\lambda)$ acting on a Hilbert space $\mathcal{H}(\lambda)$. We denote its eigenstates and eigenvalues by

$$\hat{T}(\lambda)\,|\Psi_i(\lambda)\rangle = \tau_i\,|\Psi_i(\lambda)\rangle, \qquad \tau_i \in \mathbb{R}^+,$$

with the project's assignment of $\tau_i$ to Standard-Model fermions and neutrinos listed in the Appendix.

### 2.3 Topological constraints: Berry phase and Euler-Berry condition

Given a Berry connection $A(T)$ on the parameter bundle over $M_{\text{time}}$, the Berry phase for a closed cycle $\Gamma$ is

$$\gamma_\Gamma = \oint_\Gamma A(T) \cdot dT,$$

and the project imposes a global Euler–Berry selection rule of the schematic form

$$\sum_k e^{i\gamma_{C_k}} = 0.$$

For a two-cycle case the condition reduces to a relative phase condition

$$\gamma_1 - \gamma_2 \equiv \pi \pmod{2\pi},$$

which discretizes admissible cycles and suppresses continuous degeneracies.

### 2.4 Mass operator (ansatz)

We introduce a mass operator $\hat{\mathcal{M}}(\tau)$ that depends on the cycle length. The phenomenological mapping used and tested in the files takes two forms:

1. *Power-law family map*

$$m_f = \kappa_{\text{fam}}\, \tau^{\alpha_{\text{fam}}},$$

   with family-dependent constants $(\kappa_{\text{fam}}, \alpha_{\text{fam}})$ (leptons, light quarks, heavy quarks).

2. *Polynomial in $\tau$ (mass-squared)*: for a small family set we can adopt

$$m_f^2 = c_0 + c_1\tau + c_2\tau^2,$$

   which is a convenient three-parameter fit when three anchor points are available.

### 2.5 White threads and network states

Nonlocal couplings are encoded in amplitudes $\mathcal{W}_{ij}$ (the "white threads" analogue to white hole concept):

$$\mathcal{W}_{ij} = \langle \Psi_i | e^{i \oint_{C_{ij}} A_\mu dx^\mu} | \Psi_j \rangle.$$

Network states are linear + bilinear combinations

$$|\Psi_{\text{network}}\rangle = \sum_i a_i |\Psi_i\rangle + \sum_{i<j} b_{ij}\, \mathcal{W}_{ij}\, |\Psi_i\rangle |\Psi_j\rangle,$$

with normalization $\sum_i |a_i|^2 + \sum_{i<j} |b_{ij}\mathcal{W}_{ij}|^2 = 1$.

### 2.6 Effective cosmological term

The effective cosmological term (schematic form reproduced from project definitions) is

$$\Lambda_{\text{eff}}(x) = \Lambda_{\text{vac}} + \alpha\sqrt{R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}} + \beta \sum_i \partial_\mu \Phi_i \partial^\mu \Phi_i + \gamma \sum_i V_i(\Phi_i), \tag{1}$$

augmented by topology-dependent corrections mediated by $\mathcal{W}_{ij}$ and Berry phases. Equation (1) is the formal starting point for testing cosmological consequences (CMB, large-scale correlations) as described in the project.

# 3 Family-wise fitting methodology

We now present the statistical method used to extract family parameters $(\kappa, \alpha)$ from a set of $(\tau_i, m_i)$ pairs for a family.

## 3.1 Power-law fit via log transform (ordinary least squares)

Assume

$$m_i = \kappa\, \tau_i^{\alpha}.$$

Taking natural logarithms:

$$\ln m_i = \ln \kappa + \alpha \ln \tau_i.$$

Write

$$y_i = \ln m_i, \quad x_i = \ln \tau_i.$$

This is a standard linear model

$$y_i = A + \alpha x_i + \varepsilon_i, \qquad A = \ln \kappa,$$

which we solve by ordinary least squares. The normal equations yield

$$\begin{bmatrix} \hat{A} \\ \hat{\alpha} \end{bmatrix} = \left(X^\top X\right)^{-1} X^\top \boldsymbol{y}, \quad X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \end{bmatrix},$$

and $\hat{\kappa} = e^{\hat{A}}$.

This estimator minimizes $\sum_i (\ln m_i - \hat{A} - \hat{\alpha} \ln \tau_i)^2$; it is the standard, robust way to obtain a global power-law approximation.

## 3.2 Polynomial fit for mass-squared (exact solve)

If a family contains three anchor points $(\tau_j, m_j)$, one may exactly solve for parameters $c_0, c_1, c_2$ in

$$m^2 = c_0 + c_1 \tau + c_2 \tau^2,$$

by solving the linear $3 \times 3$ system

$$\begin{bmatrix} 1 & \tau_1 & \tau_1^2 \\ 1 & \tau_2 & \tau_2^2 \\ 1 & \tau_3 & \tau_3^2 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} m_1^2 \\ m_2^2 \\ m_3^2 \end{bmatrix}.$$

This is algebraic and gives an exact reproducing mapping for the chosen three anchor points.

# 4 Data used (project assignments)

The numeric eigenvalue assignments and representative observed masses are drawn from the project's tables and are reproduced here for reproducibility.

## 4.1 Eigenvalues $\tau_i$ (project assignment)

- Leptons: $\tau_e = 4,\ \tau_\mu = 1,\ \tau_\tau = 10.$
- Light quarks: $\tau_u = 0.05,\ \tau_d = 0.1,\ \tau_s = 0.4.$
- Heavy quarks: $\tau_c = 5,\ \tau_b = 10,\ \tau_t = 100.$
- Neutrinos (nominal test values): $\tau_{\nu_1} = 0.02,\ \tau_{\nu_2} = 0.05,\ \tau_{\nu_3} = 0.1.$

### 4.2 Representative observed masses (used for fitting)

We use representative PDG / common current-mass values for the fitting exercise:

$$\begin{aligned} &m_e = 0.511 \text{ MeV}, \quad m_\mu = 105.7 \text{ MeV}, \quad m_\tau = 1777 \text{ MeV}, \\ &m_u \approx 2.2 \text{ MeV}, \quad m_d \approx 4.7 \text{ MeV}, \quad m_s \approx 96 \text{ MeV}, \\ &m_c \approx 1270 \text{ MeV}, \quad m_b \approx 4180 \text{ MeV}, \quad m_t \approx 173000 \text{ MeV}. \end{aligned}$$

## 5 Solving the fits — explicit algebra and numeric results

We report both the power-law family-wise fits (via log-OLS) and a polynomial exact solve for the charged-lepton set. All numeric results are reproducible from the data above.

### 5.1 Power-law fits (log-OLS) per family

Using the method of Section 3, we compute $\hat{\kappa}$ and $\hat{\alpha}$ for each family.

**Leptons.** Data: $(\tau, m)$ pairs $(4, 0.511), (1, 105.7), (10, 1777)$. Log-OLS yields

$$\hat{\kappa}_{\text{lep}} \approx 16.791369, \qquad \hat{\alpha}_{\text{lep}} \approx 0.8157778.$$

Predicted masses (MeV): [52.03, 16.79, 109.87] for $(e, \mu, \tau)$ respectively. Percent errors relative to the listed observed masses are large for this fit (see Appendix table). Conclusion: a single global power-law of this form is not a compact representation for the three leptons with the chosen $\tau$ assignment — a polynomial mass-squared fit is more appropriate here (see below).

**Light quarks (u,d,s).** Log-OLS fit gives

$$\hat{\kappa}_{\text{LQ}} \approx 477.370393, \qquad \hat{\alpha}_{\text{LQ}} \approx 1.8672955.$$

Predicted masses (MeV): [1.776, 6.480, 86.255] for $(u, d, s)$. Percent residuals are modest for $u$ and $s$, larger for $d$; light-quark current masses are scale-dependent and therefore naturally less precise.

**Heavy quarks (c,b,t).** Log-OLS fit yields

$$\hat{\kappa}_{\text{HQ}} \approx 93.829892, \qquad \hat{\alpha}_{\text{HQ}} \approx 1.6345097.$$

Predicted masses (MeV): [1302.61, 4044.37, 174325.40] for $(c, b, t)$. Percent residuals are $\lesssim 3\%$ across the heavy family, showing a good compact family-wise power-law description.

**Comment.** The heavy-quark family admits an excellent two-parameter power-law representation with the chosen $\tau$ values; the leptons are better represented by a three-parameter polynomial in $\tau$ when anchoring to three points.

### 5.2 Polynomial fit for charged leptons (exact solve)

We solve for coefficients $c_0, c_1, c_2$ in

$$m^2 = c_0 + c_1\tau + c_2\tau^2$$

using $(\tau, m) = (4, 0.511), (1, 105.7), (10, 1777)$. Solving the linear system yields:

$$\begin{aligned} c_0 &\approx 2.504575438218 \times 10^5 \text{ MeV}^2, \\ c_1 &\approx -2.981752982038 \times 10^5 \text{ MeV}^2, \\ c_2 &\approx 5.889024438217 \times 10^4 \text{ MeV}^2. \end{aligned}$$

This reproduces the charged-lepton masses exactly (by construction) and provides a mass-squared polynomial operator that maps the three anchor $\tau$ values to their observed masses. The coefficients are recorded in the Appendix.

### 5.3 Neutrino splitting test (pairwise scaling)

The project suggests mapping neutrino mass-squared differences via $\Delta m_{ij}^2 \propto (\tau_i^2 - \tau_j^2)$ up to a scale factor and topology correction:

$$\Delta m_{ij}^2 = S\left(\tau_i^2 - \tau_j^2\right) \cdot F_{ij}(\mathcal{W}_{ij}, \gamma_i, \chi_i),$$

where $F_{ij}$ encodes topological/pairwise corrections (default $F_{ij} = 1$ is the simplest hypothesis).

Using $\tau_\nu = (0.02, 0.05, 0.1)$ and observed $\Delta m_{21}^2 \approx 7.5 \times 10^{-5}$ eV$^2$, solve for $S$ under $F_{21} = 1$:

$$S = \frac{\Delta m_{21}^2}{\tau_2^2 - \tau_1^2} = \frac{7.5 \times 10^{-5}}{0.05^2 - 0.02^2} \approx 0.0357143.$$

Predicting $\Delta m_{31}^2$ with the same $S$ (and $F_{31} = 1$) gives

$$\Delta m^{2\,\text{pred}}_{31} = S\big(0.1^2 - 0.02^2\big) \approx 3.43 \times 10^{-4} \text{ eV}^2,$$

while the observed value is $\approx 2.4 \times 10^{-3}$ eV$^2$. The simple global-scaling hypothesis $F_{ij} \equiv 1$ fails to reproduce both measured splittings simultaneously. This demonstrates the need for pairwise or topology-dependent correction factors $F_{ij}$ derived from $\mathcal{W}_{ij}$ and Berry/Euler data — a falsifiable statement: if a unique topology-derived $F_{ij}$ cannot be produced, the model's neutrino mapping fails.

## 6 Emergence of Discrete Mass Spectra via Topological Time-Loops

In the standard formulation of the Standard Model, fermion masses are treated as free parameters derived from Yukawa couplings to the Higgs field. In the *Metatime* framework, we propose that mass is essentially an emergent property resulting from the topological stability of closed time loops.

The effective Lagrangian density for a fermion field $\psi$ interacting with the metatime background field $\mathcal{T} = (\phi, T_{\mu\nu})$ is given by:

$$\mathcal{L}_{eff} = \bar{\psi}\left(i\gamma^\mu \mathcal{D}_\mu - \hat{\mathcal{M}}(\tau)\right)\psi \tag{2}$$

Here, the scalar mass term $m$ is replaced by the **Mass Operator** $\hat{\mathcal{M}}(\tau)$, which is functionally dependent on the discrete path length $\tau$ of the particle's trajectory within the Collatz-like configuration space. Based on our numerical family-wise fits, the eigenvalue of this operator follows a distinct power-law scaling:

$$m_f = \langle \psi_f | \hat{\mathcal{M}}(\tau) | \psi_f \rangle = \kappa_{fam} \cdot (\tau_{cycle})^{\alpha_{fam}} \tag{3}$$

where:

- $\tau_{cycle}$ represents the integer length of the closed Collatz orbit (analogous to the "rhythm" of the cycle).
- $\kappa_{fam}$ and $\alpha_{fam}$ are topological constants specific to the fermion family (e.g., for the lepton sector, $\alpha \approx 2.97$).

The stability of these states is governed by the global **Euler-Berry Condition**. A quantum state is physically realizable (on-shell) only if the accumulated Berry phase $\gamma_{\mathcal{T}}$ along the closed time-loop trajectory $\Gamma$ satisfies the resonance condition:

$$1 + e^{i \oint_\Gamma \mathcal{A}_\tau d\tau} = 0 \quad \Longrightarrow \quad \gamma_{\mathcal{T}} = \pi(2n+1) \tag{4}$$

This topological constraint forces the otherwise continuous spectrum of possible energies to collapse into discrete "shells" (observed particle masses), effectively quantizing the dynamic interaction between the particle and the metatime manifold.

# 7 Geometric Quantization of Shells via the Metatime Lagrangian

In the Metatime framework, we redefine the concept of atomic shells not as static spatial regions, but as topological attractors within the metatime manifold $\mathcal{M}_{time}$. The observed stability of electrons on specific shells is a direct consequence of arithmetic resonance.

The effective Lagrangian for a lepton $\psi$ in this background is formulated as:

$$\mathcal{L} = \bar{\psi} \left[ \gamma^\mu (i\partial_\mu - g\mathcal{A}_\mu) - \kappa \hat{\mathcal{T}}^\alpha \right] \psi \tag{5}$$

In this expression, the mass term is replaced by the **Time-Cycle Operator** $\hat{\mathcal{T}}$, whose eigenvalues are discrete integers $\tau$ derived from the underlying Collatz-like topology of the time field. For the lepton family, we utilize the fit $\kappa \approx 0.511$ and $\alpha \approx 2.97$.

The "dance" of the electron is physically allowed (on-shell) only when the action integral over a closed metatime trajectory $\Gamma$ satisfies the phase-locking condition:

$$\mathcal{S}_{eff} = \oint_\Gamma P_\tau \, d\tau \equiv \pi(2n+1) \pmod{2\pi} \tag{6}$$

This implies that the electron's position on a shell is a manifestation of the *Euler-Berry condition*. When the simple arithmetic of the cycle length $\tau$ aligns with the global curvature of the time field, the Lagrangian reaches a local minimum, creating the stable "standing wave" of strings of matter that we observe as an atom.

# 8 Cosmological consequences and falsifiable targets

From the effective $\Lambda_{\text{eff}}(x)$ (Eq. 1) and the coupling of metatime to standard fields via $\mathcal{W}_{ij}$ and Berry curvature, the project identifies explicit falsifiable predictions:

1. **Neutrino CP resonance signatures:** topology-induced shifts in $\delta_{\text{CP}}$ that can be tested by DUNE and Hyper-K.
2. **CMB low-$\ell$ anomalies:** a topology-modulated enhancement of large-angle power (multipoles $\ell \lesssim 100$) associated with metatime perturbations, testable against Planck/ACT and future Stage-4 CMB datasets.
3. **Galaxy correlation modulations:** scale-dependent modulations at $\mathcal{O}(10^2\text{Mpc})$ manifested as small deviations from standard 2-point correlation predictions (Euclid/DESI).
4. **Neutrino-induced fine structure of $\Lambda_0(x)$:** correlation between neutrino oscillation phases and local departures of $\Lambda_{\text{eff}}$ — testable (in principle) with joint neutrino + cosmology analyses (IceCube + large-scale surveys).

Each prediction is falsifiable: if the predicted topology-dependent signature is absent at the sensitivity claimed in the project (and declared tolerances), that specific model instantiation is rejected.

## 9 Discussion and model limitations

- The model is a *constructive* framework: it provides explicit operators, data assignments, and fitting procedures that turn topology + metatime into testable spectra. That is its scientific strength.
- The neutrino sector is the clear open point: a naive global scaling fails and requires topology-derived pairwise corrections. This is not a bug but a focal prediction (the model predicts where standard fits will fail).
- All conclusions depend on the numeric $\tau_i$ assignments and the chosen observable masses. Sensitivity tests should be run if alternative $\tau_i$ assignments are proposed.
- Cosmological consequences require full dynamical simulation (perturbed FRW with $\Lambda_{\text{eff}}(x)$ coupled to metatime); this manuscript provides the formal skeleton and a list of falsifiable signatures. Numerical simulation is the next necessary step.

## 10 Conclusion

We provide a fully explicit metatime cosmological specification with algebraic solves for the particle mapping, an explicit neutrino-splitting test that fails under the simplest hypothesis (thereby making a falsifiable demand on topology corrections), and a list of cosmological signatures that can be compared to current and near-future experiments. The document is intentionally a calculation and testing package: all fits, coefficients and residuals are stated, and all key model assumptions are declared so that the theory can be accepted, refined, or falsified by data.

## A Reproducibility notes

All numerical solves reported here are simple linear algebra and ordinary least-squares on the explicit $(\tau, m)$ sets listed in Section 4. The neutrino scaling test is algebraic. The project files contain pseudocode and full numerical tables; this manuscript uses exactly the project $\tau$ assignments and the representative masses listed above. Any alternative $\tau$ assignments will change the numeric outputs; the algebraic procedures remain identical.

**Acknowledgements.** This technical document packages the Metatime project's cosmological specifications into a reproducible calculation and falsification suite. The work intentionally separates hypotheses, calculations and falsifiable targets.

## B Numeric solutions and full tables

Table 1: Family-wise power-law fits (log-OLS). Predicted masses and percent errors.

| Family | $\hat{\kappa}$ | $\hat{\alpha}$ | Particle | $\tau$ | Predicted $m$ (MeV) | % error vs. observed |
|---|---|---|---|---|---|---|
| Leptons | 16.791369 | 0.8157778 | $e$ | 4 | 52.03 | +10081.5% |
| | | | $\mu$ | 1 | 16.79 | -84.11% |
| | | | $\tau$ | 10 | 109.87 | -93.82% |
| Light quarks | 477.370393 | 1.8672955 | $u$ | 0.05 | 1.776 | -19.27% |
| | | | $d$ | 0.10 | 6.480 | +37.87% |
| | | | $s$ | 0.40 | 86.255 | -10.15% |
| Heavy quarks | 93.829892 | 1.6345097 | $c$ | 5 | 1302.61 | +2.57% |
| | | | $b$ | 10 | 4044.37 | -3.24% |
| | | | $t$ | 100 | 174325.40 | +0.77% |

Table 2: Exact polynomial fit for charged leptons: $m^2 = c_0 + c_1\tau + c_2\tau^2$.

| Coefficient | Value (MeV$^2$) | Notes |
|---|---|---|
| $c_0$ | $2.504575438218 \times 10^5$ | constant term |
| $c_1$ | $-2.981752982038 \times 10^5$ | linear in $\tau$ |
| $c_2$ | $5.889024438217 \times 10^4$ | quadratic in $\tau$ |

Table 3: Neutrino scaling test.

| Pair | $\Delta(\tau^2)$ | Predicted $\Delta m^2$ under $F_{ij} = 1$ |
|---|---|---|
| (2,1): $\tau_2 = 0.05, \tau_1 = 0.02$ | 0.0021 | By fit $\Delta m^2_{21} = 7.5 \times 10^{-5}$ eV$^2$ (used to fix $S$) |
| (3,1): $\tau_3 = 0.1, \tau_1 = 0.02$ | 0.0096 | $\Delta m^{2\,\mathrm{pred}}_{31} = S \times 0.0096 \approx 3.43 \times 10^{-4}$ eV$^2$ |
| | | (observed $\Delta m^2_{31} \approx 2.4 \times 10^{-3}$ eV$^2$) |